

AI Management: Cost, KPIs & ROI

Detailed further in Sprint 9 (stories 9.1, 9.2, 9.3). These are the working numbers now.

AI Management: Cost, KPIs & ROI

Cost Analysis

Per-diagnosis compute cost

Item	Cost driver	Estimated cost per diagnosis
Use Case Readiness (OpenAI, reasoning: high, web search enabled)	Input + output tokens, typically 15k-40k tokens per run	EUR 0.15 to 0.45
Fix Router (OpenAI, reasoning: medium, web search enabled)	Smaller prompt, fewer output tokens	EUR 0.05 to 0.15
LLM-as-Judge (OpenAI, reasoning: medium, no web search)	No search cost, moderate output	EUR 0.03 to 0.08
Platform Risk (6 parallel DeepSeek calls + Pinecone queries)	Cheaper model, but more calls per report	EUR 0.10 to 0.25
Pinecone queries (incident + framework corpus)	Per-query cost is small but scales with usage	Under EUR 0.01 per query

Full diagnostic cycle (diagnosis + Fix Router + Judge), worst case: roughly EUR 0.70 per client engagement. This is the number that matters for pricing, not the individual line items.

Monthly hosting cost

Service	Plan	Monthly cost
Vercel (Groundwork frontend)	Hobby (current), moving to Pro (needed for 6-module production scale)	EUR 0, moving to roughly EUR 20/seat
Railway (Platform Risk agent)	Usage-based	roughly EUR 15 to 40, depending on volume
Pinecone	Starter/Standard	EUR 0 to 70, depending on index size at 6-module scale
OpenAI / DeepSeek API usage	Usage-based, see per-diagnosis table above	Scales directly with client volume

Known gap, flagged honestly: the Vercel Hobby plan’s hard 300-second function limit (discovered and worked around in Round 2) will not hold at production scale across six modules. Upgrading to Pro before Sprint 10 (Vercel launch) is a real, required line item, not optional.

Total estimated monthly operating cost

At BrainyFuture’s scale (10 consultants, roughly 15 diagnoses/consultant/month):
150 diagnoses/month x roughly EUR 0.70 full-cycle cost = **roughly EUR 105/month in AI compute**, plus **roughly EUR 55 to 130/month in hosting**, once at full 6-module production scale.
Total: roughly EUR 160 to 235/month to operate Groundwork across all 10 consultants. This is a real, low number relative to consulting fees, worth stating plainly in the business case.

KPI Framework

KPI	What it measures	Data source	Target
	% of client engagements where		

Module adoption rate	% of client engagements where a Groundwork module is actually used	Report creation logs (Vercel Blob), grouped by module	80%+ of engagements by end of Sprint 10
Average diagnosis turnaround time	Time from intake submission to completed report	Timestamps already logged in each report's createdAt field	Under 3 minutes, 90th percentile
Judge agreement rate	% of Judge reviews returning "agrees" with zero significant findings	JudgeReview.agreement field, aggregated	70%+ (a lower rate isn't necessarily bad, it means the Judge is catching real things, but it's worth tracking as a quality signal)
Fix Router acceptance rate	% of Fix Router suggestions a consultant actually acts on	Requires a new lightweight tracking field, not yet built, this is itself a Sprint 4/5 follow-up	40%+
Report reuse / share rate	% of reports where "Copy share link" or "Download PDF" is used at least once	ReportActions.tsx click events, needs basic analytics added	60%+
Significant-finding escalation rate	% of reports with a Judge "significant" finding that get the Sprint 4.5 acknowledgment workflow triggered	Escalation log (Sprint 4.5 dependency)	100% (this one should always be 100%, it's a compliance safeguard, not a growth metric)

Honest note: two of these KPIs (Fix Router acceptance, report reuse) need small tracking additions that don't exist yet. That's itself worth naming to the jury, a real KPI framework should say what's measurable today versus what needs light instrumentation first, not pretend everything is already tracked.

Platform-Wide ROI

Assumptions

Assumption	Value	Basis
BrainyFuture consultants	10	Stated capstone scenario
Diagnoses per consultant per month	roughly 15	Estimated, one every 1-2 client touchpoints
Manual time to do this work without Groundwork	roughly 90 minutes per use case (research + risk assessment + write-up)	Consultant judgment, unverified estimate
Time with Groundwork (review + edit AI output)	roughly 20 minutes per use case	Estimated from Round 2 demo timings
Consultant loaded hourly cost	EUR 45/hour	EU consulting role, salary + overhead
Total build cost (Round 1 + Round 2 + this roadmap)	roughly EUR 35,000	See breakdown below

Build cost breakdown

Phase	Estimated cost
Round 1 (Use Case Readiness MVP, dashboard, n8n POC)	roughly EUR 8,000
Round 2 (jurisdiction fix, performance fix, Incident-Grounded Evidence, Fix Router, LLM-as-Judge)	roughly EUR 9,000
Sprints 4-11 (remaining UCR items, 3 new modules, business layer, integration)	roughly EUR 18,000
Total	roughly EUR 35,000

Value calculation

Time saved per diagnosis: 90 minus 20 = 70 minutes = 1.17 hours
Value per diagnosis: 1.17 hours x EUR 45 = **EUR 52.65 saved per diagnosis**

150 diagnoses/month x EUR 52.65 = **EUR 7,897.50/month in consultant time saved**
EUR 7,897.50 x 12 = **EUR 94,770/year**

94,770/year

ROI

12 months: Total cost = EUR 35,000 (build) + (EUR 200/month x 12) (operating, mid-estimate) = EUR 37,400 Net benefit = EUR 94,770 minus EUR 37,400 = EUR 57,370 ROI = $(57,370 / 37,400) \times 100 = \mathbf{153\%}$

36 months: Total cost = EUR 35,000 + (EUR 200 x 36) = EUR 42,200 Total value = EUR 94,770 x 3 = EUR 284,310 Net benefit = EUR 284,310 minus EUR 42,200 = EUR 242,110 ROI = $(242,110 / 42,200) \times 100 = \mathbf{574\%}$

Break-even

EUR 35,000 build cost / EUR 7,897.50 monthly value is approximately **4.4 months**, faster than Jane's individual-use-case break-even (roughly 8 months), because this is spread across all 10 consultants at once, not one client engagement.

Every number above with a stated basis is an estimate, not measured data. The KPI framework above is what turns these into real, tracked numbers by the end of Sprint 9. This page should be revisited and replaced with actuals once real usage data exists, that revisit is itself Sprint 9's deliverable.